

Solved Question Paper

BCS-011: Computer Basics and PC Software
IGNOU BCA Term-End Examination, June 2022
Time: 3 hours Maximum Marks: 100

This solution booklet gives clean, exam-ready answers for all questions in the June 2022 BCS-011 paper. Diagrams and flowcharts are redrawn in TikZ for clarity.

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Question 1

1(a) Perform the following operations.

Answer.

(i) Convert $(235)_{10}$ to binary.

$$235 = 128 + 64 + 32 + 8 + 2 + 1.$$

Therefore,

$$(235)_{10} = (11101011)_2.$$

(ii) Convert $(10.3750)_{10}$ to binary.

$$10_{10} = 1010_2, \quad 0.375 \times 2 = 0.75 \Rightarrow 0,$$

$$0.75 \times 2 = 1.5 \Rightarrow 1, \quad 0.5 \times 2 = 1.0 \Rightarrow 1.$$

Therefore,

$$(10.3750)_{10} = (1010.011)_2.$$

(iii) Convert $(1011101)_2$ to decimal.

$$(1011101)_2 = 1 \times 64 + 0 \times 32 + 1 \times 16 + 1 \times 8 + 1 \times 4 + 0 \times 2 + 1 = 93.$$

Thus,

$$(1011101)_2 = (93)_{10}.$$

(iv) Convert $(FABC)_{16}$ to binary.

$$F = 1111, \quad A = 1010, \quad B = 1011, \quad C = 1100.$$

Therefore,

$$(FABC)_{16} = (1111 \ 1010 \ 1011 \ 1100)_2.$$

(v) Convert $(A5)_{16}$ to decimal.

$$(A5)_{16} = 10 \times 16^1 + 5 \times 16^0 = 160 + 5 = 165.$$

Therefore,

$$(A5)_{16} = (165)_{10}.$$

1(b) What will be the storage capacity of the disk pack?

Answer. A disk pack has 4 plates. Since each plate has two usable recording surfaces, the total number of recording surfaces is

$$4 \times 2 = 8.$$

Given:

$$\text{Tracks per surface} = 2048, \quad \text{Sectors per track} = 512,$$

$$\text{Bytes per sector} = 1024.$$

Hence,

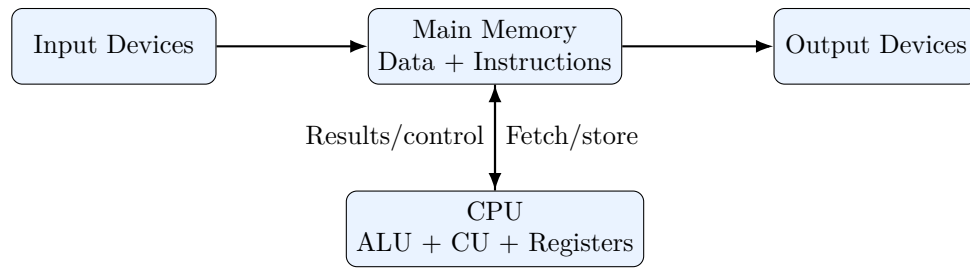
$$\text{Capacity} = 8 \times 2048 \times 512 \times 1024 \text{ bytes.}$$

$$= 8,589,934,592 \text{ bytes} = 8 \text{ GiB approximately.}$$

Thus the storage capacity is about **8 GB** using binary gigabytes (GiB) convention.

1(c) Explain the von Neumann Architecture with the help of a diagram.

Answer. The von Neumann architecture is a stored-program computer architecture in which program instructions and data are stored in the same main memory. The CPU fetches instructions from memory, decodes them, executes them, and stores results back in memory or sends them to output devices.



Main components are: input unit, main memory, CPU, output unit, and buses. The CPU contains the Arithmetic Logic Unit, Control Unit, and registers. The main feature is that both data and instructions share the same memory and bus system.

1(d) Explain two features of the software needed for each operation.

Answer.

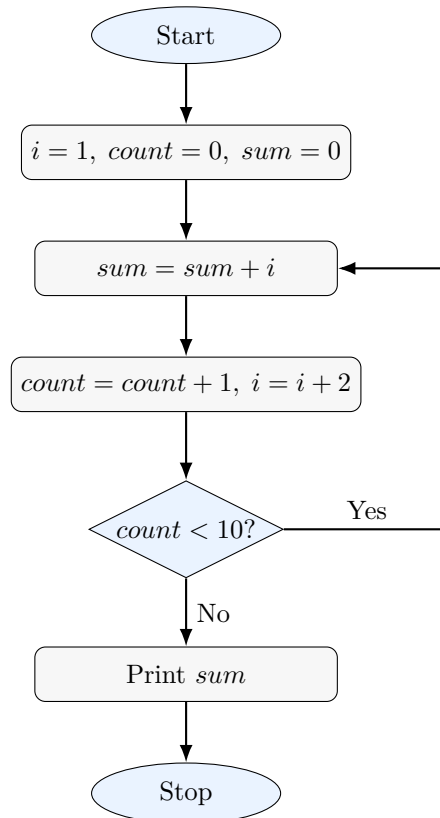
Operation	Required software	Two features
Keeping track of files and directories	Operating system / file manager	Creates, renames, copies, moves and deletes files; maintains directory structure, file paths, permissions and storage locations.
Translating high-level language program to machine language	Compiler / language translator	Checks syntax and reports errors; converts complete high-level source code into object or executable machine code.
Identifying and removing computer virus	Antivirus software	Scans files, memory and storage for malware signatures or suspicious behaviour; quarantines, deletes or repairs infected files.

1(e) Write an algorithm and draw the flowchart for finding the sum of the first 10 odd numbers.

Answer. The first 10 odd numbers are 1, 3, 5, ..., 19. Their sum is 100, but the algorithm below computes it step by step.

Algorithm

1. Start.
2. Set $i = 1$, $count = 0$, $sum = 0$.
3. Add i to sum .
4. Increase $count$ by 1.
5. Increase i by 2 to get the next odd number.
6. If $count < 10$, repeat from step 3.
7. Print sum .
8. Stop.



1(f) Given the IP addresses and subnet masks, find the Net-ID.

Answer. The Net-ID is obtained by applying bitwise AND between the IP address and the subnet mask.

IP address	Subnet mask	Net-ID
10.1.5.7	255.0.0.0	10.0.0.0
192.168.1.05	255.255.255.0	192.168.1.0

1(g) Explain the purpose of the following networking devices.

Answer.

NIC: A Network Interface Card connects a computer to a network. It provides a physical port or wireless interface and has a unique MAC address.

Modem: A modem modulates and demodulates signals. It converts digital computer data into a form suitable for transmission over telephone/cable/fiber links and converts received signals back into digital form.

Repeater: A repeater regenerates weak or distorted network signals so that they can travel longer distances. It works at the physical layer and does not filter traffic.

1(h) List any six advantages of e-learning.

Answer. Six advantages are:

1. Learners can study from any location with Internet access.
2. It supports self-paced learning and repeated viewing of lessons.
3. It reduces travel, classroom and printed-material costs.
4. It can use multimedia such as video, animation, quizzes and simulations.

5. Progress can be tracked through tests, assignments and learning analytics.
6. It allows quick updating and distribution of course content.

1(i) List two advantages and one disadvantage of using Inkjet Printer.

Answer. Advantages:

- Inkjet printers produce good-quality colour images and photographs.
- They are usually inexpensive, compact and suitable for home or small-office use.

Disadvantage: Ink cartridges may be costly and prints may smudge or fade if low-quality ink or paper is used.

Question 2

2(a) Define instruction and explain the role of Control Unit in execution.

Answer. An instruction is a binary command that tells the processor what operation to perform and on which data. A typical instruction contains an opcode and operand information.

Example: In a symbolic instruction such as `ADD R1, R2`, the opcode is `ADD`, and the operands are registers `R1` and `R2`. It tells the CPU to add the contents of the registers.

The Control Unit performs the following roles during instruction execution:

1. Fetches the instruction from memory using the Program Counter.
2. Decodes the opcode to identify the operation.
3. Generates control signals for ALU, memory and registers.
4. Coordinates data movement between registers, memory and ALU.
5. Updates the Program Counter for the next instruction.

Thus, the Control Unit directs and coordinates the execution of instructions.

2(b) Differentiate between the following.

Answer. (i) Static RAM and Dynamic RAM

SRAM	DRAM
Uses flip-flop circuits.	Uses capacitors to store bits.
Faster and costlier.	Slower and cheaper.
Does not need periodic refresh.	Needs periodic refresh.
Used mainly in cache memory.	Used mainly as main memory.

(ii) RAM and ROM

RAM	ROM
Read/write memory.	Mostly read-only memory.
Volatile; contents are lost when power is off.	Non-volatile; contents remain after power is off.
Stores programs and data currently in use.	Stores firmware or boot instructions.

(iii) Hard disk and Magnetic tapes

Hard disk	Magnetic tape
Direct/random access storage.	Sequential access storage.
Useful for frequently used data and operating systems.	Useful for backup and archival storage.
Faster retrieval of individual files.	Slower retrieval because tape must be wound to the required location.

(iv) DVD and CD

DVD	CD
Higher storage capacity, commonly 4.7 GB or more.	Lower storage capacity, commonly around 700 MB.
Uses shorter wavelength laser and denser tracks.	Uses less dense storage pattern.
Better for video and large software distribution.	Better for audio and smaller data files.

2(c) What is the role of a port in a computer? Explain parallel and serial ports.

Answer. A port is an interface through which a computer communicates with external devices such as printers, keyboards, modems, storage devices and scanners. Ports provide physical connection points and communication rules.

Parallel port: A parallel port transfers multiple bits simultaneously through several wires. It was commonly used for printers. It can be faster over short distances but requires more wires and is less suitable for long-distance communication.

Serial port: A serial port transfers data one bit at a time. It was commonly used for modems, mice and some communication devices. It requires fewer wires and is more reliable over longer distances, though older serial ports were slower than parallel ports.

Question 3

3(a) Explain client/server architecture and three-tier client/server architecture.

Answer. In client/server architecture, a **client** is a computer or program that requests a service, and a **server** is a computer or program that provides that service. For example, a web browser is a client and a web server provides web pages.

In three-tier architecture, the system is divided into three layers: presentation layer, application/business logic layer and database layer.



Advantages include better scalability, easier maintenance, improved security and separation of user interface from database operations.

3(b) What is open-source software? Objectives and examples.

Answer. Open-source software is software whose source code is available for users to view, use, modify and distribute under an open-source licence.

Objectives of the open-source movement:

- To encourage collaborative development and peer review.
- To reduce dependence on proprietary software and promote freedom to use, study, modify and share software.

Open-source software	Purpose
Linux	Operating system
LibreOffice	Office suite for documents, spreadsheets and presentations
Mozilla Firefox	Web browser
GIMP	Image editing software

3(c) Define the following in the context of an Operating System.

Answer.

Batch processing: Jobs are collected and executed in batches without continuous user interaction. It is useful for payroll, billing and report generation.

Graphical User Interface: A GUI allows users to interact with the computer using windows, icons, menus, buttons and a pointer instead of only text commands.

Multiprogramming: Several programs are kept in memory at the same time so that the CPU can switch to another program when one waits for I/O, improving CPU utilisation.

Time-sharing system: CPU time is divided into small time slices and shared among many users or processes, giving each user an interactive response.

3(d) Output of C expressions for $x = 5$, $y = 8$, $z = 13$.

Answer. In C, relational expressions return 1 for true and 0 for false.

Expression	Evaluation	Output
$x == y$	$5 == 8$ is false	0
$(x + y) == z$	$5 + 8 = 13$, so $13 == 13$ is true	1
$y < z$	$8 < 13$ is true	1
$x != z$	$5 \neq 13$ is true	1

Question 4

4(a) Explain the terms in the context of Computer Network.

Answer.

Half duplex: Communication is possible in both directions, but not at the same time. Example: walkie-talkie communication.

Channel: A communication path or medium through which data travels from sender to receiver, such as cable, optical fibre or radio link.

Packet switching: Data is divided into small packets. Each packet may travel independently through the network and is reassembled at the destination.

Bandwidth: The data-carrying capacity of a communication channel, usually measured in bits per second. Higher bandwidth means more data can be transmitted per unit time.

4(b) Explain characteristics of LAN. How are LANs different from WANs?

Answer. A Local Area Network connects computers and devices within a limited area such as a room, building, school or campus.

Characteristics of LAN:

- Covers a small geographical area.
- Usually owned and managed by one organisation.
- Provides high data-transfer speed and low error rate.
- Allows sharing of files, printers, Internet connection and applications.
- Commonly uses Ethernet or Wi-Fi.

LAN	WAN
Covers a small area such as office or campus.	Covers large areas such as cities, countries or continents.
Usually faster and cheaper to maintain.	Usually slower and costlier because it uses telecom links.
Owned by one organisation. Example: college lab network.	May be owned by multiple service providers. Example: Internet or bank network across cities.

4(c) What is a URL? Explain DNS with example.

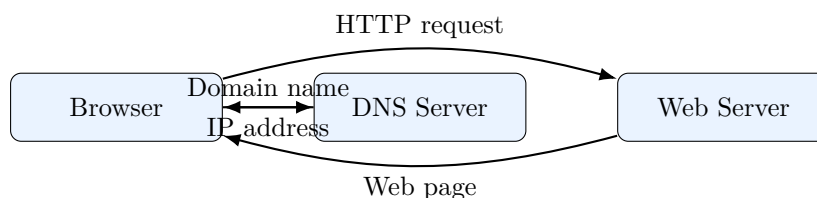
Answer. A URL (Uniform Resource Locator) is the complete address used to locate a resource on the Internet. It identifies the protocol, domain name and path of a resource.

Example:

`https://www.ignou.ac.in/studentzone`

Here `https` is the protocol, `www.ignou.ac.in` is the domain name and `/studentzone` is the path.

DNS (Domain Name System) translates human-readable domain names into IP addresses. For example, when a user enters `www.ignou.ac.in`, DNS finds the corresponding IP address of the web server. The browser then uses that IP address to contact the server.



4(d) Explain characteristics of Wiki and Social Networking.

Answer. Wiki:

- A collaborative website where users can create, edit and link pages.
- Maintains revision history so old versions can be recovered.
- Useful for documentation, knowledge bases, class notes and project collaboration.
- Supports collective editing and quick updating of information.

Social networking:

- Allows users to create profiles and connect with other users.
- Supports sharing of text, photos, videos, links and messages.
- Enables groups, communities, discussions and public or private communication.
- Useful for communication, marketing, professional networking and social interaction.

Question 5

5(a) Security Threats on Internet

Answer. Internet security threats are risks that can damage data, privacy, devices or services. Common threats include viruses, worms, Trojans, spyware, phishing, identity theft, spam, ransomware and denial-of-service attacks. Protection methods include antivirus software, firewall, strong passwords, two-factor authentication, regular updates, safe browsing and backup.

5(b) TCP/IP Model

Answer. The TCP/IP model explains how data is prepared, addressed, transmitted and received over networks including the Internet.

Application Layer: HTTP, FTP, SMTP, DNS
Transport Layer: TCP, UDP
Internet Layer: IP, ICMP
Network Access Layer: Ethernet, Wi-Fi

Application layer provides services to user applications. Transport layer provides end-to-end delivery. Internet layer handles logical addressing and routing using IP. Network access layer transmits frames over physical media.

5(c) Object-Oriented Programming Language

Answer. An object-oriented programming language organises programs around objects. An object combines data and methods that operate on that data. Main features include class, object, encapsulation, inheritance, polymorphism and abstraction.

Example: A class **Student** may have data such as roll number and marks, and methods such as `display()` or `calculateGrade()`. C++, Java and Python support object-oriented programming.

5(d) Perverse Software

Answer. Perverse software is malicious or harmful software designed to disturb, damage, steal or misuse computer resources. It includes viruses, worms, Trojan horses, spyware, adware and ransomware. It may delete files, steal passwords, monitor users, encrypt data or reduce system performance. Antivirus, firewall, updates and cautious downloading help reduce the risk.

5(e) Subroutine and Functions

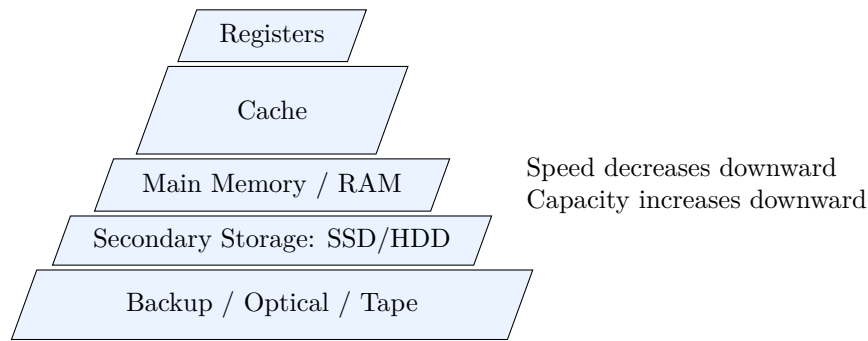
Answer. A subroutine is a named block of code used to perform a specific task. It is called whenever that task is required, which reduces repetition.

A function is similar to a subroutine but usually returns a value to the calling program. For example, a function `sum(a,b)` may return `a+b`. A subroutine `printReport()` may only print a report and may not return a value.

Subroutine	Function
Performs a task.	Performs a task and commonly returns a value.
May not be used inside expressions.	Can be used inside expressions if it returns a value.
Example: print student list.	Example: calculate average marks.

5(f) Memory Hierarchy

Answer. Memory hierarchy arranges storage according to speed, cost and capacity. Faster memories are costlier and smaller; slower memories are cheaper and larger.



5(g) Classification of Printers

Answer. Printers may be classified as impact and non-impact printers.

Impact printers strike an ink ribbon against paper. Examples: dot matrix and daisy wheel printers. They are noisy but can print multipart forms.

Non-impact printers print without physical striking. Examples: inkjet and laser printers. They are quieter and give better print quality.

They may also be classified by print quality and use: character printers, line printers and page printers; monochrome and colour printers; photo printers and office printers.

5(h) Features of Spreadsheet Software

Answer. Spreadsheet software is used to store, calculate, analyse and present data in rows and columns. Important features include:

- Cell-based data entry in rows and columns.
- Formulas and functions for automatic calculation.
- Charts and graphs for visual presentation.
- Sorting, filtering and conditional formatting.
- What-if analysis, tables and pivot-style summaries.
- Workbook with multiple worksheets.

Examples include Microsoft Excel, LibreOffice Calc and Google Sheets.